Kushali Darak | Data Engineer - AI Data Engineer

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PROFESSIONAL SUMMARY

• AI Data Engineer with **4+ years of experience building scalable batch and real-time data pipelines** using Python, SQL, PySpark, and Apache Spark across manufacturing and financial services environments.

• Experienced in designing **distributed data processing workflows using Databricks, Apache Airflow, and Kafka**, enabling reliable ingestion, transformation, and streaming data delivery for enterprise analytics platforms.

• Hands-on experience working with **AWS cloud services (S3, Glue, Lambda, Redshift) and Snowflake data warehouses** to build and optimize ETL/ELT pipelines supporting high-volume analytics and reporting workloads.

• Skilled in developing **feature engineering pipelines, ML dataset preparation workflows, and MLflow-tracked data pipelines** to support machine learning model experimentation and MLOps collaboration with data science teams.

• Strong background in **data validation, schema validation, and data quality frameworks such as Great Expectations**, ensuring reliable and production-ready data pipelines across distributed systems.

• Comfortable collaborating in **Agile engineering environments**, working closely with analytics, machine learning, and platform teams to deliver reliable and scalable data infrastructure.

SKILLS

Programming Languages: Python, SQL, Scala
Data Engineering & Big Data: Apache Spark, PySpark, Databricks, ETL/ELT Pipeline Development, Data Transformation, Distributed Data Processing, Batch Processing, Workflow Orchestration
Streaming & Event Processing: Apache Kafka, Spark Structured Streaming, Event-Driven Data Pipelines, Real-Time Data Processing
Cloud Platforms: Amazon Web Services (AWS S3, AWS Glue, Amazon Redshift, AWS Lambda, EC2), Microsoft Azure (Azure Data Factory, Azure Synapse Analytics), Google Cloud Platform (BigQuery)
Data Warehousing & Databases: Snowflake, Amazon Redshift, PostgreSQL, MySQL, MongoDB, Dimensional Modeling, Data Lakehouse Architecture
Pipeline & Orchestration Tools: Apache Airflow, dbt
AI / Machine Learning Data Engineering: Feature Engineering, Data Preprocessing, ML Dataset Preparation, MLflow, Model Data Versioning, Data Quality Validation, MLOps Support
Data Analysis & Visualization: Power BI, Tableau, Pandas, NumPy
Data Formats & APIs: JSON, Parquet, Avro, REST APIs
DevOps & Infrastructure: Git, GitHub, GitLab, Docker, CI/CD Pipelines, Linux, Cloud Logging, SLA Alerting, Data Pipeline Health Monitoring
Data Quality & Testing: Great Expectations, Data Validation Frameworks, Schema Validation, Unit Testing for Data Pipelines
Methodologies: Agile, Scrum, SDLC, Cross-Functional Collaboration, Problem Solving, Analytical Thinking

WORK EXPERIENCE

AI Data Engineer | Rockwell Automation – Wisconsin, USA **Aug 2024 – Feb 2026**

- Designed and implemented scalable **PySpark ETL pipelines on Databricks** in collaboration with data platform engineers and **integrated with AWS S3 and AWS Glue**, processing high-volume manufacturing telemetry streams exceeding millions of events daily and reducing ingestion latency by 30%.
- Contributed to building **Kafka-based real-time streaming pipelines using Spark Structured Streaming**, enabling event-driven processing of equipment telemetry signals and supporting predictive maintenance analytics across manufacturing operations.
- Optimized **Snowflake warehouse transformations using dbt models orchestrated via Apache Airflow**, improving query performance for datasets exceeding **300M+ records** and accelerating downstream analytics and machine learning dataset preparation.
- Developed **Python-based feature engineering and ML dataset preparation pipelines integrated with MLflow**, enabling reproducible data versioning and improved collaboration with ML engineers building predictive models.
- Integrated REST APIs with **AWS Glue batch jobs and AWS Lambda triggers** to build automated ingestion workflows processing millions of telemetry records daily within a scalable data lakehouse architecture.
- Containerized data processing workflows using **Docker** for consistent deployment across environments.
- Implemented **data validation and schema validation checks using Great Expectations and pipeline testing practices**, improving reliability and quality of datasets consumed by analytics and machine learning teams.

Data Engineer | HCL Technologies – India **Sep 2022 – Nov 2023**

- Developed **distributed Spark SQL and PySpark pipelines on Databricks** to transform high-volume financial transaction datasets, improving batch pipeline throughput by 24% while supporting enterprise reporting workloads.
- Built **Apache Airflow DAG orchestration pipelines** for automated ingestion into **AWS Redshift and Snowflake data warehouse environments**, improving data availability and pipeline visibility for analytics teams.
- Designed **dimensional data models and incremental loading strategies** supporting Power BI reporting dashboards, improving reporting reliability and data consistency for financial analytics workflows.
- Refactored legacy **Python ETL pipelines into modular PySpark transformations**, improving scalability and reducing compute resource consumption across distributed data processing environments.
- Monitored production pipelines using **cloud logging, pipeline monitoring, and SLA alerting mechanisms**, ensuring reliable data delivery and operational stability across analytics data platforms.

Associate Data Engineer | Cognizant – India **Apr 2021 – Aug 2022**

- Built **SQL-based ETL ingestion pipelines** migrating structured datasets from on-premise systems into **AWS S3 data lake environments**, improving accessibility of enterprise analytics datasets.
- Supported **data preprocessing and transformation workflows using Python Pandas and NumPy**, enabling structured ML dataset preparation and improving training data readiness for analytics teams.
- Assisted implementation of **Kafka event ingestion pipelines** capturing customer interaction data streams, enabling near real-time Tableau dashboards for operational reporting.
- Implemented **data reconciliation scripts, validation checks, and unit testing within ETL pipelines**, improving data quality and reducing reporting inconsistencies across downstream analytics systems.
- Supported development and testing of ETL workflows by validating source-to-target data mappings and performing reconciliation checks across staging and warehouse datasets.

EDUCATION

Master of Science in Computer Science
Concordia University Wisconsin (CUW), Wisconsin, USA

CERTIFICATIONS

- **Microsoft Certified: Azure Data Engineer Associate (DP-203)**
- **Databricks Certified Data Engineer Associate**

PROJECTS

Real-Time Predictive Maintenance Data Platform

- Designed end-to-end streaming pipelines using Kafka and Spark Structured Streaming to process manufacturing telemetry events and enable predictive maintenance analytics dashboards.
- Built scalable PySpark transformation workflows on Databricks integrated with AWS S3 storage, improving ingestion throughput by nearly 28% across monitored production systems.
- Developed feature engineering datasets tracked using MLflow experiments to support machine learning model training and ensure reproducible data preparation workflows.
- Automated Airflow orchestration for batch and streaming dependencies, reducing pipeline failure recovery time by approximately 22% and improving operational stability.

Enterprise Financial Data Warehouse Optimization

- Engineered distributed Spark SQL ETL pipelines transforming high-volume financial transaction datasets into Snowflake warehouse layers supporting enterprise reporting and analytics consumption.
- Implemented incremental loading and dimensional modeling strategies improving Power BI dashboard refresh reliability and reducing data latency by nearly 19%.
- Optimized warehouse compute utilization by refactoring legacy Python batch scripts into scalable PySpark workflows deployed on Databricks clusters environments.

Cloud-Based Machine Learning Dataset Engineering Framework

- Developed automated data ingestion pipelines integrating REST APIs with AWS Glue jobs and Lambda triggers to streamline machine learning dataset preparation workflows.
- Designed preprocessing modules using Python Pandas and NumPy libraries reducing manual cleansing effort by approximately 21% for analytics teams.
- Implemented data versioning and experiment tracking using MLflow integrated storage improving model reproducibility and collaboration across distributed data science environments.
- Enabled near real-time analytics delivery by integrating Kafka event ingestion streams supporting Tableau dashboards monitoring customer behavior performance metrics.